

PRAKTIKUM SISTEM OPERASI

MODUL 9

Syscall



Oleh:

Martryatus Sofia

2311104003

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

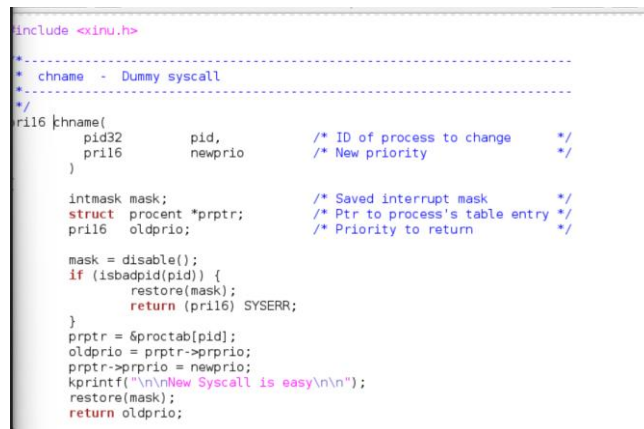
UNIVERSITAS TELKOM

2026

JURNAL

1. Soal 1

[50 Poin] Buat syscall baru seperti yang ditunjukkan pada modul syscall poin 9.5!
(sertakan Screenshot kode dan hasil run)



```
include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----*/

r16 kname(
    pid32    pid,      /* ID of process to change */
    pri16    newprio   /* New priority */
)
{
    intmask mask;      /* Saved interrupt mask */
    struct procent *prptr; /* Ptr to process's table entry */
    pri16    oldprio;   /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16) SYSERR;
    }
    prptr = &proctab[pid];
    oldprio = prptr->prprio;
    prptr->prprio = newprio;
    kprintf("\n\nNew Syscall is easy\n\n");
    restore(mask);
    return oldprio;
}
```

2. Soal 2

[25 Poin] Perbaiki syscall chprio (xinu/system/chprio.c) dengan memperhatikan validasi input

- Pastikan id adalah angka dari 0 – NPROC (ukuran maks banyaknya proses)
- Pastikan prioritas adalah bilangan yang positif
- Compile dan jalankan Xinu dengan syscall yang telah diperbaiki
- make clean
- make

```

/*
 * chprio - Change the scheduling priority of a process
 */
syscall chprio(
    pid32    pid,          /* ID of process to change */
    prlp6    newprio       /* New priority */
)
{
    intmask mask;          /* Saved interrupt mask */
    struct procnt *prptr;   /* Ptr to process's table entry */
    prlp6    oldprio;       /* Priority to return */

    mask = disable();
    if (pid < 0 || pid >= NPROC) {
        restore(mask);
        return SYSERR;
    }
    prptr = &proctab[pid];
    if (prptr->prstate == PR_FREE || newprio <= 0) {
        restore(mask);
        return SYSERR;
    }
    oldprio = prptr->prprio;
    prptr->prprio = newprio;

    restore(mask);
    return oldprio;
}

```

```

/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION="" "cat version\" -I../i
nclude -o binaries/clkdir.o ../system/clkdir.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION="" "cat version\" -I../i
nclude -o binaries/ctxsw.o ../system/ctxsw.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION="" "cat version\" -I../i
nclude -o binaries/intr.o ../system/intr.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION="" "cat version\" -I../i
nclude -o binaries/ttydispatch.o ../device/tty/ttydispatch.S
/usr/bin/gcc-4.8 -march=i586 -m32 -fno-builtin -fno-stack-protector -nostdlib -c -Wall
-00 -DBRELOC=0x150000 -DBOOTPLOC=0x150000 -DBSDURG -DVERSION="" "cat version\" -I../i
nclude -o binaries/ethdispatch.o ../device/eth/ethdispatch.S
Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.
xinu@xinu-develop-end:~/xinu/compile$ cd ..
xinu@xinu-develop-end:~/xinu$ cd system
xinu@xinu-develop-end:~/xinu/system$ gedit chprio.c
xinu@xinu-develop-end:~/xinu/system$

```

3. Soal 3

Lakukan hal-hal berikut ini

- Edit xsh_uptime.c Tambahkan kode berikut

```

/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

// tambahan code
chprio(5,33) // memanggil syscall chprio; mengubah prioritas menjadi 33 pada id=5

```

-
- Compile source code tersebut dengan perintah
- make clean
- make

- Jalankan perintah ps
- xsh \$ ps
- perhatikan prioritas proses dengan id = 5
- Jalankan uptime
- xsh \$ uptime
- Perhatikan hasil perintah tersebut
- Jalankan ps
- xsh \$ ps
- perhatikan prioritas proses dengan id = 5 seharusnya sudah berubah

```

if (nargs == 2 && strcmp(args[1], "--help", 7) == 0) {
    kprintf("Use: %s\n\n", args[0]);
    kprintf("Description:\n");
    kprintf("\tDisplays time since the system booted\n");
    kprintf("Options:\n");
    kprintf("\t--help\t display this help and exit\n");
    return 0;
}

/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

chprio(5,33);
secs = clktime;

days = secs/secperday;
secs -= days*secperday;

hrs = secs/secperhr;
secs -= hrs*secperhr;

```

```

-----NAMA-----
Martryatus Sofia
-----NIM-----
2311104003
-----

Welcome to Xinu Martryatus Sofia!

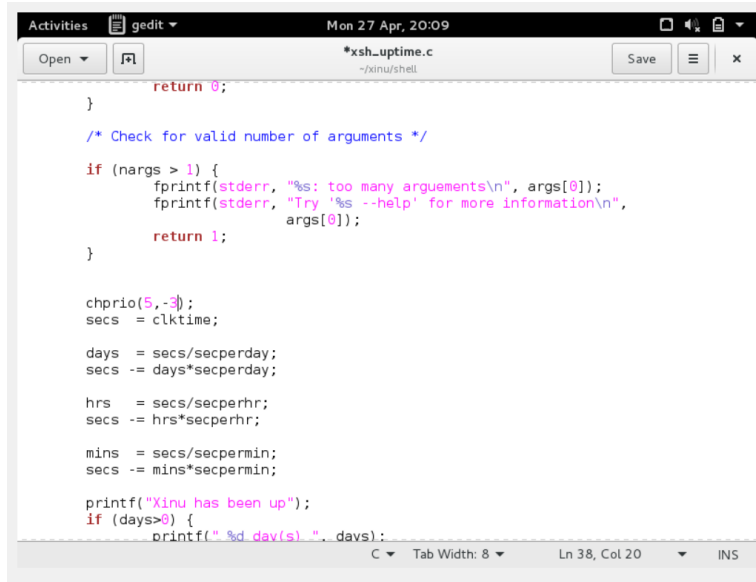
xsh $ ps
command (xinu) not found
xsh $ uptime
Xinu has been up 15 second(s)
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv   33   4 0x005D7FFC 0x005D79AC 8192 0 6
7 ps              curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0

```

4. Soal 4

Testing chprio syscall yang telah diubah

- Testing prioritas tidak boleh < 0 : Ubah “chprio(5,33)” menjadi “chprio(5,-3)” pada xsh_uptime.c



```
return 0;

/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

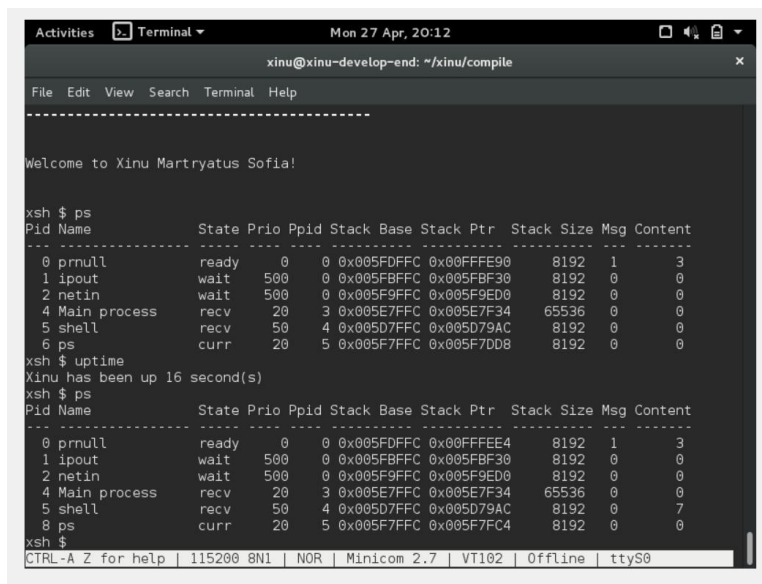
chprio(5,-3);
secs = clktime;

days = secs/secperday;
secs -= days*secperday;

hrs = secs/secperhr;
secs -= hrs*secperhr;

mins = secs/secpermin;
secs -= mins*secpermin;

printf("Xinu has been up");
if (days>0) {
    printf(" %d day(s) ", days);
```



```
xinu@xinu-develop-end: ~/xinu/compile

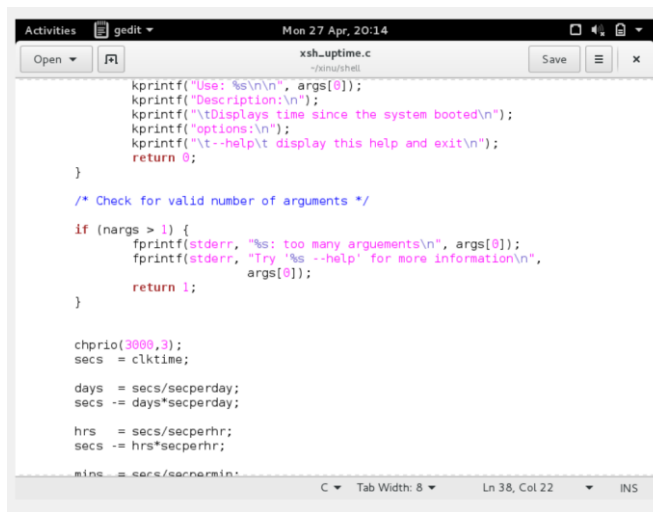
Welcome to Xinu Martryatus Sofia!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE90 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps         curr   20   5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $ uptime
Xinu has been up 16 second(s)
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps         curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

- Testing id adalah valid: Ubah “chprio(5,33)” menjadi “chprio(3000,3)”



```
Activities gedit Mon 27 Apr, 20:14
xsh_uptime.c
-jenushell

kprintf("Use: %s\n", args[0]);
kprintf("Description:\n");
kprintf("\tDisplays time since the system booted\n");
kprintf("options:\n");
kprintf("\t--help\t display this help and exit\n");
return 0;
}

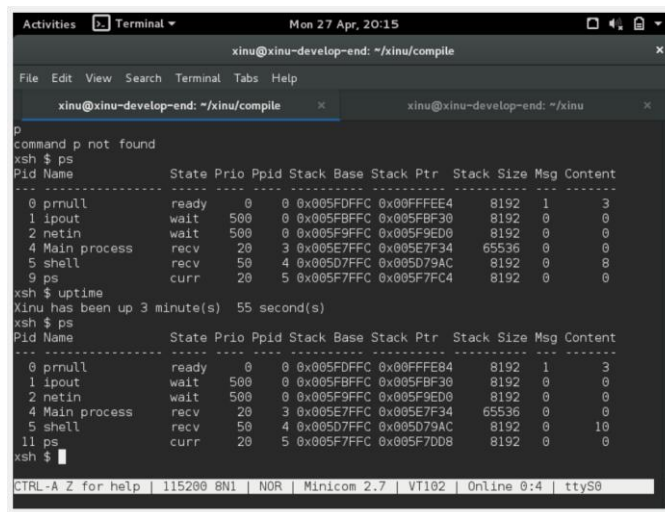
/* Check for valid number of arguments */
if (nargs > 1) {
    fprintf(stderr, "%s: too many arguments\n", args[0]);
    fprintf(stderr, "Try '%s --help' for more information\n",
            args[0]);
    return 1;
}

chprio(3000, 3);
secs = cktime;

days = secs/secperday;
secs -= days*secperday;

hrs = secs/secperhr;
secs -= hrs*secperhr;

mins = secs/secpermin;
```



```
Activities Terminal Mon 27 Apr, 20:15
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/compile xinu@xinu-develop-end: ~/xinu

p
command p not found
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 8
9 ps         curr   20   5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $ uptime
Xinu has been up 3 minute(s) 55 second(s)
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE84 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005D7FFC 0x005D79AC 8192 0 10
11 ps        curr   20   5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:4 | ttyS0
```

- Hasil dua testing di atas adalah prioritas tidak berubah karena salah argument (dibuktikan dengan menggunakan perintah ps)

Iya tidak ada yang berubah

```
Activities Terminal Mon 27 Apr, 20:12
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-----
Welcome to Xinu Martryatus Sofia!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE90 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process  recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps         curr  20   5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $ uptime
Xinu has been up 16 second(s)

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process  recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps         curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

```
Activities Terminal Mon 27 Apr, 20:15
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Tabs Help
xinu@xinu-develop-end: ~/xinu/compile xinu@xinu-develop-end: ~/xinu
p
command p not found
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process  recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005D7FFC 0x005D79AC 8192 0 8
9 ps         curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $ uptime
Xinu has been up 3 minute(s) 55 second(s)

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE84 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process  recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv  50   4 0x005D7FFC 0x005D79AC 8192 0 10
11 ps        curr  20   5 0x005F7FFC 0x005F7DD8 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:4 | ttyS0
```